REPORT
DEX2013

By:

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Problem Statement

A machine operated automatically need to be monitored by factory personnel. Without attention, the machine can malfunction . If this issue goes unsolved, this will heavily damage the machine and render it unrepairable. This will incur monetary loss for the factory. So, a solution is required as soon as possible.

Objective

To create a program that classifies machine readings into different statuses.

Inputs, Processing and Outputs (IPO)

Input	Process	Output
• Temperature • Pressure • Vibration • Current • RPM	• Check temperature, pressure, vibration, current and RPM according to classification rules to determine status	• Status

Classification tests

Test	Temperature(°C)	Pressure(bar)	Vibration(mm/s)	Current(A)	RPM	Result (Status)
1.	35	4.0	2.0	3.5	1450	NORMAL
2.	40	4.0	2.0	3.5	1450	WARNING
3.	50	4.0	2.0	3.5	1450	WARNING
4.	50.1	4.0	2.0	3.5	1450	CRITICAL
5.	35	6.1	2.0	3.5	1450	CRITICAL
6.	35	4.0	6.1	3.5	1450	CRITICAL
7.	35	4.0	2.0	5.1	1450	CRITICAL
8.	35	4.0	2.0	3.5	1349	CRITICAL
9.	35	4.0	2.0	3.5	1390	WARNING
10.	35	1.9	2.0	3.5	1450	WARNING

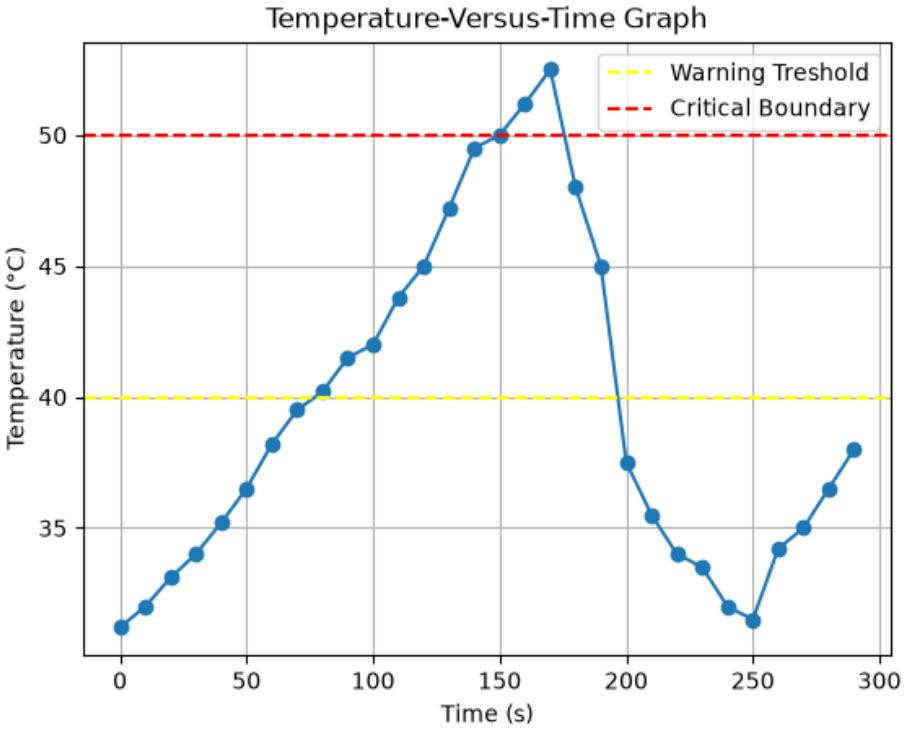
Program Output

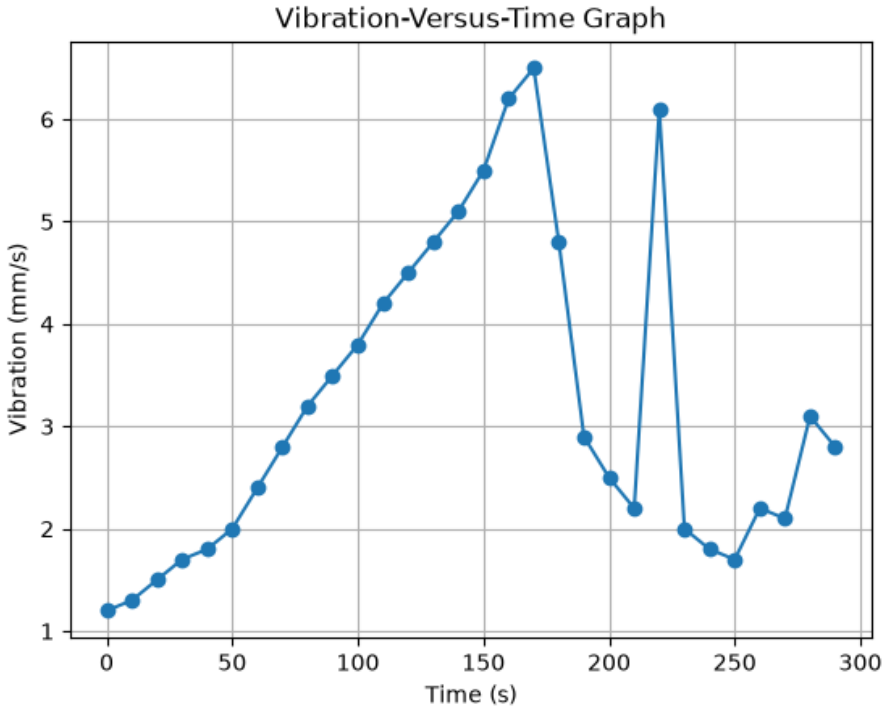
```
PS C:\Users\eeson\Documents\DEX2013> & C:\Users\eeson\AppData\Local\Programs\Python\Python314\python.exe c:/Users/eeson/Documents/DEX2013/Assignment.py
The CSV file has been succesfully loaded.
   Reading_ID  Time_s  Temperature_C  Pressure_bar  Vibration_mm_s  Motor_Current_A   RPM
0           1       0           31.2           4.1             1.2              3.1  1480
1           2      10           32.0           4.0             1.3              3.2  1478
2           3      20           33.1           4.1             1.5              3.3  1482
3           4      30           34.0           3.9             1.7              3.4  1475
4           5      40           35.2           4.0             1.8              3.5  1470
Number of columns = 7
Number of rows = 30
Mean Temperature = 39.44 °C
Mean Pressure = 3.56 bar
Mean Vibration = 3.21 mm/s
Mean Current = 4.04 A
Mean RPM = 1447.03 RPM
Max Temperature = 52.5 °C
Max Pressure = 6.2 bar
Max Vibration = 6.5 mm/s
Max Current = 5.4 A
Max RPM = 1580.0 RPM
Min Temperature = 31.2 °C
Min Pressure = 1.7 bar
Min Vibration = 1.2 mm/s
Min Current = 3.1 A
Min RPM = 1320.0 RPM
No. of NORMAL = 10
No. of WARNING = 13
No. of CRITICAL = 7
% NORMAL = 33.33 %
% WARNING = 43.33 %
% CRITICAL = 23.33 %
```

	Reading_ID	Time_s	Temperature_C	Pressure_bar	Vibration_mm_s	Motor_Current_A	RPM	Status	Recommended_Action
0	1	0	31.2	4.1	1.2	3.1	1480	NORMAL	Continue normal monitoring in the simulation.
1	2	10	32.0	4.0	1.3	3.2	1478	NORMAL	Continue normal monitoring in the simulation.
2	3	20	33.1	4.1	1.5	3.3	1482	NORMAL	Continue normal monitoring in the simulation.
3	4	30	34.0	3.9	1.7	3.4	1475	NORMAL	Continue normal monitoring in the simulation.
4	5	40	35.2	4.0	1.8	3.5	1470	NORMAL	Continue normal monitoring in the simulation.
5	6	50	36.5	3.8	2.0	3.6	1468	NORMAL	Continue normal monitoring in the simulation.
6	7	60	38.2	3.7	2.4	3.8	1460	NORMAL	Continue normal monitoring in the simulation.
7	8	70	39.5	3.5	2.8	3.9	1455	NORMAL	Continue normal monitoring in the simulation.
8	9	80	40.2	3.4	3.2	4.0	1450	WARNING	Continue monitoring and arrange inspection.
9	10	90	41.5	3.3	3.5	4.1	1445	WARNING	Continue monitoring and arrange inspection.
10	11	100	42.0	3.2	3.8	4.2	1438	WARNING	Continue monitoring and arrange inspection.
11	12	110	43.8	3.1	4.2	4.3	1435	WARNING	Continue monitoring and arrange inspection.
12	13	120	45.0	3.0	4.5	4.4	1425	WARNING	Continue monitoring and arrange inspection.
13	14	130	47.2	2.9	4.8	4.6	1410	WARNING	Continue monitoring and arrange inspection.
14	15	140	49.5	2.8	5.1	4.8	1405	WARNING	Continue monitoring and arrange inspection.
15	16	150	50.0	2.7	5.5	5.0	1395	WARNING	Continue monitoring and arrange inspection.
16	17	160	51.2	2.6	6.2	5.2	1380	CRITICAL	Stop operation in the simulation and request supervisor/technician inspection.
17	18	170	52.5	2.5	6.5	5.4	1360	CRITICAL	Stop operation in the simulation and request supervisor/technician inspection.
18	19	180	48.0	1.8	4.8	4.4	1410	WARNING	Continue monitoring and arrange inspection.
19	20	190	45.0	4.0	2.9	3.9	1460	WARNING	Continue monitoring and arrange inspection.
20	21	200	37.5	4.2	2.5	3.7	1470	NORMAL	Continue normal monitoring in the simulation.
21	22	210	35.5	6.2	2.2	3.6	1480	CRITICAL	Stop operation in the simulation and request supervisor/technician inspection.
22	23	220	34.0	4.0	6.1	4.3	1485	CRITICAL	Stop operation in the simulation and request supervisor/technician inspection.
23	24	230	33.5	4.0	2.0	5.3	1475	CRITICAL	Stop operation in the simulation and request supervisor/technician inspection.
24	25	240	32.0	4.1	1.8	3.5	1320	CRITICAL	Stop operation in the simulation and request supervisor/technician inspection.
25	26	250	31.5	4.0	1.7	3.4	1580	CRITICAL	Stop operation in the simulation and request supervisor/technician inspection.
26	27	260	34.2	4.2	2.2	3.8	1390	WARNING	Continue monitoring and arrange inspection.
27	28	270	35.0	1.7	2.1	3.7	1460	WARNING	Continue monitoring and arrange inspection.
28	29	280	36.5	4.1	3.1	3.8	1480	WARNING	Continue monitoring and arrange inspection.
29	30	290	38.0	4.0	2.8	3.9	1490	NORMAL	Continue normal monitoring in the simulation.

(Full data with added columns)

Graphs


Temperature-Versus-Time Graph
Warning Treshold
Critical Boundary
Temperature (°C)
50
45
40
35
0
50
100
150
200
250
300
Time (s)


Vibration-Versus-Time Graph
Vibration (mm/s)
6
5
4
3
2
1
0
50
100
150
200
250
300
Time (s)

Counts and Percentages

Status	Count	Percentage
NORMAL	10	33.33%
WARNING	13	43.33%
CRITICAL	7	23.33%

Trends and Unusual Readings

1. Vibration increases when temperature increases.
2. No. of WARNING is more than NORMAL
3. The vibration on second 220 is unusually high

Recommendations

Continue monitoring and look out for abnormalities while also conduct weekly maintenance for smoother operation of the machine.

End of report